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A three-dimensional gravity inversion applied to São Miguel Island (Azores)

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Abstract. Gravimetric studies are becoming more and more widely acknowledged as a useful tool for studying and modeling the distributions of subsurface masses that are associated with volcanic activity. In this paper, new gravimetric data for the volcanic island of São Miguel (Azores) were analyzed and interpreted by a stabilized linear inversion methodology. An inversion model of higher resolution was calculated for the Caldera of Furnas, which has a larger density of data. In order to filter out the noncorrelatable anomalies, least squares prediction was used, resulting in a correlated gravimetric signal model with an accuracy of the order of 0.9 mGal. The gravimetric inversion technique is based on the adjustment of a three-dimensional (3-D) model of cubes of unknown density that represents the island's subsurface. The problem of non-uniqueness is solved by minimization with appropriate covariance matrices of the data (resulting from the least squares prediction) and of the unknowns. We also propose a criterion for choosing a balance between the data fit (which in this case corresponds to residues with rms of the order of 0.6 mGal) and the smoothness of the solution. The global model of the island includes a low-density zone in a WNW-ESE direction and a depth of the order of 20 km, associated with the Terceira rift spreading center. The minimums located at a depth of 4 km may be associated with shallow magmatic chambers beneath the main volcanoes of the island. The main high-density area is related to the Nordeste basaltic shield. With regard to the Caldera Furnas, in addition to the minimum that can be associated with a magmatic chamber, there are other shallow minimums that correspond to eruptive processes.

Introduction

São Miguel is the main island of the Azores archipelago. It is located at 37.8°N, 25.5°W, in the eastern group of the archipelago. The origin of these volcanic islands is connected with the triple junction of the North American, the African, and the Eurasian plates. The third branch, the Eurasian/African plate boundary, of this triple junction includes the Azores spreading center, which comprises a series of "en echelon" rifted basins running from the Formigas Trough at 36.8°N, 24.5°W to a point near 39.3°N, 28.8°W [*Searle*, 1980] and which were collectively called the Terceira Rift (Figures 1 and 2). This spreading center trends 120° across the archipelago and shows recent volcanic and seismic activity, concentrated in a series of volcanic ridges and massifs which flank the older rifted basins (Figure 2). The Terceira Rift basins were possibly formed by rifting of the preexisting Atlantic lithosphere first, and then after the plate was ruptured, the spreading center evolved into a more typical hot spot ridge structure with the spreading axis associated with a series of en echelon axial volcanic ridges which now flank the rifted basins.

Close to São Miguel, the main structures connected to the spreading axis are São Miguel Basin (SE of the island), São Miguel Island, and the South Hirondelle Basin (NW of the island). The basins tend to have a square cross section and to include well-developed scarps trending about 142°-154°. The ridges that flank the basins are also characterized by faults with similar trends to those in the basins walls but of smaller amplitude [*Searle*, 1980]. In the case of the ridge running northwest from São Miguel, there are suggestions of the presence of recently active NW-SE fault systems, such as the Mosteiros Graben on the flank of Sete Cidades and the Congro Graben on the flank of the Agua de Pau Volcano [*Guest et al.*, 1994], where the NW-SE structural features end in a WSW-ENE trend. The existence of several historical eruptions (Sete Cidades 1444, Picos 1652, Fogo 1563, Furnas 1630, 1440) and current intense geothermal phenomena, point to the permanent volcanic activity (Figure 3).

On São Miguel Island, the area of Furnas has been selected as a laboratory volcano within the European Environment Program. Furnas is the easternmost of the three active stratovolcanoes on the island. It is constructed on the southwestern flank of an old dissected mainly lava complex of Povoação/Nordeste. Unlike the other central volcanoes, Furnas has not built a distinctive cone but consists of a steep-sided caldera complex 7 km x 4.5 km across, which is the result of several distinct collapses. Throughout most of its history, Furnas appears to have been an essentially explosive volcano, erupting materials of trachytic composition and showing eruptions associated with large-scale caldera collapse followed by a series of caldera-

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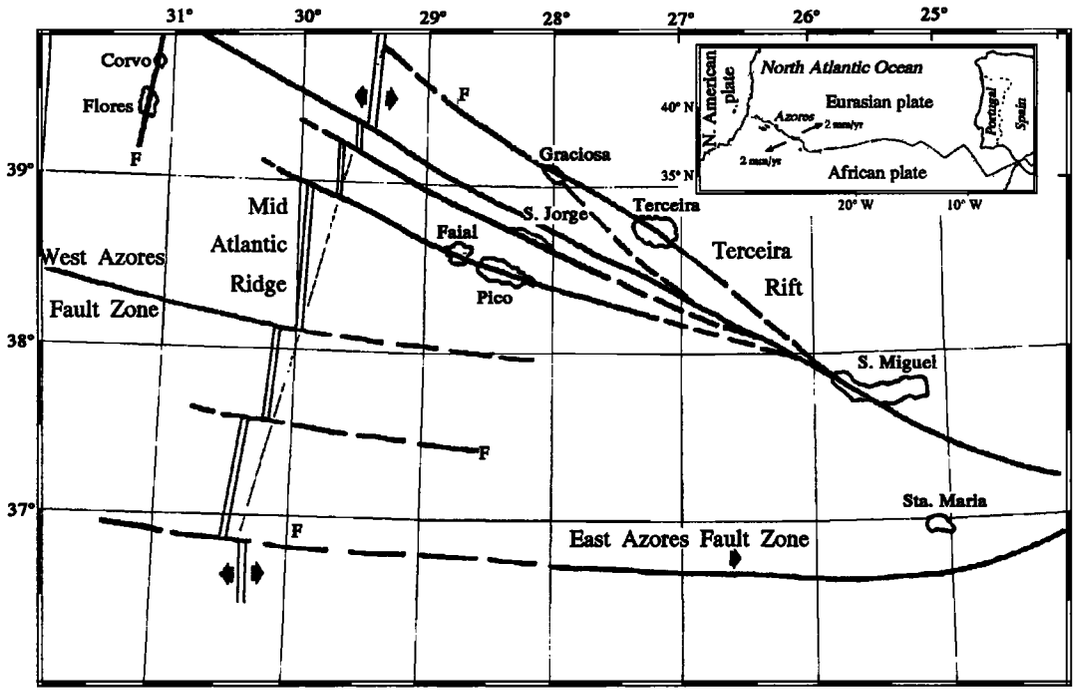
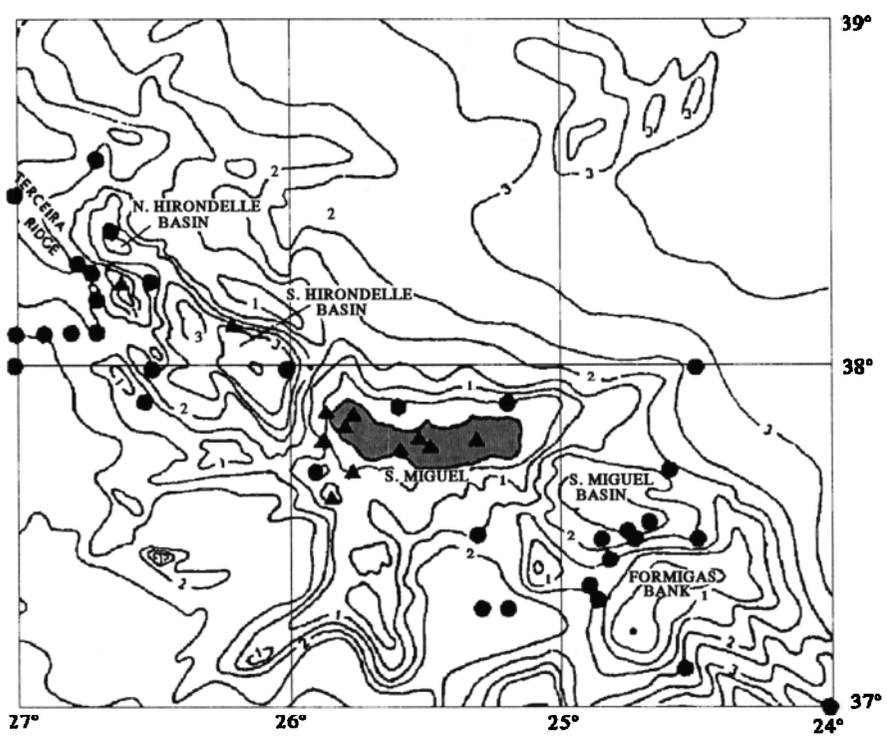


Figure 1. Location of the Azores connected to the triple junction of the North American, the African, and the Eurasian plates. Tectonic sketch of faults distribution (adapted from *Forjaz* [1984]).

filling eruptions [*Guest et al.*, 1994]. In topographic terms, the remains of three successive calderas may be observed (Figure 3). The NE, north, and SE edges of the oldest caldera are visible, whereas in the south it may have been downfaulted by a series of N-S normal faults forming a graben. Cutting the infill of the old caldera, we will see a second caldera inside it that extends the old caldera toward the east and forms a new slope to the north. During the infill



▲ Historical volcanic eruption ● Earthquake epicentres (1948-1977)

Figure 2. Bathymetry in kilometers. Contour interval 500 m. Close to São Miguel, the Terceira Rift is composed of volcanic ridges and older rifted basins (reprinted from *Searle* [1980] with kind permission of Elsevier Science - NL, Sara Burgerhartstraat 25, 1055 KV Amsterdam, The Netherlands).).

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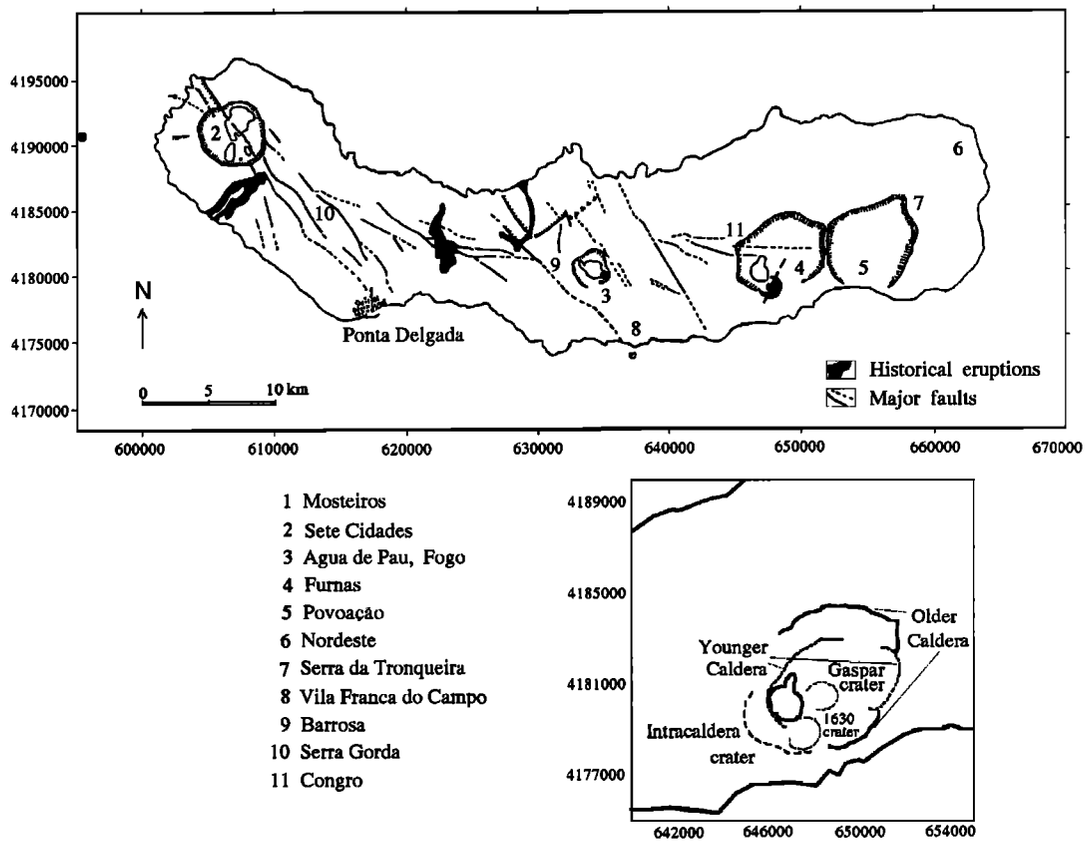

Figure 3. Volcanic sketch of São Miguel showing the locations of the calderas with their lakes, the major faults, and historical eruptions (adapted from *Forjaz* [1986]). Lower figure shows location of current crater and caldera walls associated to the Caldera de Furnas.

phases and the ensuing collapse, the caldera spread to the SW in an intracaldera zone. The two historic eruptions that took place in Furnas date back to 1440, associated with the Pico do Gaspar dome to the east of the lake, and to 1630 to the south of the caldera. Although the volcano has not erupted since 1630, the area is subject to seismicity, and the presence of a vigorous hydrothermal system indicates the existence of a heat source below the caldera at the present time. Recent Global Positioning System (GPS) and leveling investigations [*Sigmundsson et al.*, 1995] suggest an inflation in the northern part of the Caldera with a central uplift rate of 7 mm/yr, which may be caused by accumulation of magma at a shallow depth beneath the caldera.

To investigate the inner mass density distribution of the island and particularly of the Furnas region, we conducted a gravimetric study composed of 1122 stations throughout the island, with major coverage of the central and Furnas regions. The reduction of these new data gave us a gravity anomaly map from which a density contrast model is calculated by gravimetric inversion.

Solving the inverse gravimetric problem involves determining a mathematical operator to act directly on observed gravity data and to return a subsurface density distribution that can reproduce the observed field. Such an interpretation must be considered as nonunique for two reasons: First, there is a fundamental nonuniqueness in potential field inversion, and second, gravity data are generally measured only on discrete points and are subject to experimental errors. For the first problem, the challenge is to develop criteria that select solutions that are geologically possible and reasonable. These may be obtained by providing sufficient prior information as geological constraints and by choosing the appropriate minimization criterion. The second problem has a suitable treatment within information theory [*Tarantola*, 1987], especially for the general linear inverse problem.

Within the gravity inversion methods, we can distinguish between the discrete methods (adopting a finite number of parameters) and the functional methods such as, for instance, the collocation approach [*Barzaghi et al.*, 1992] or the spectral approach [*Xia and Sprowl*, 1992; *Parker*, 1973]. Another interesting aspect of the gravity inversion methods is the type of unknown parameters. First, those that adopt fixed density contrast values and look for geometrical determinations [*Richardson and MacInnes*, 1989; *Knudsen and Strykowski*, 1993; *Zhelev*, 1994] must work with nonlinear functions. Second, those of fixed geometry and unknown density contrast values can apply an immediate linear inversion [*Bear et al.*, 1995; *Burkhard and Jackson*, 1976]. There are other approaches which account for concepts of information theory and which are able to determine geometric and density parameters simultaneously [*Tarantola*, 1987; *Strykowski*, 1995].

In this paper, we propose a linear 3-D discrete inversion, taking as unknowns the densities of a distribution of prisms that reproduce the volume of the subsoil. The problem of nonuniqueness is solved by a mixed criterion of minimization of the adjustment residues and of the differences between the estimated solutions and the initial solution,

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determined by the covariance matrices and a control parameter. Moreover, the problem of the gravimetric inversion instability, caused by very local anomalies and noise, is reduced by prior treatment with least squares prediction. This also gives us a regular distribution of data that facilitates the inversion process. The density contrast model that we calculated for the island, and in particular for Furnas, is consistent with known structural tendencies and with volcanic episodes that have taken place in the area, but it also reveals new characteristics of the area that translate into anomalous density contrasts.

Gravity Data and Topographic Model

We have considered two different types of gravity data: The first were observed within this study by the Instituto de Astronomía y Geodesia (IAG, Spain) and the second were published by the Instituto Nacional de Meteorologia e Geofisica, Portugal [*Mendes and Gonçalves*, 1980]. The Portuguese gravity data correspond to 813 stations that cover the central region of São Miguel Island. The observations were carried out during 1977 with a WORDEN-255 gravimeter of 0.01 mGal counter reading accuracy. The Instituto de Astronomía y Geodesia carried out two gravimetric campaigns in 1993 and 1994, collecting data from 309 stations. The instrument used was the La Coste-Romberg G665 gravimeter, with feedback system and electronic reading, giving a reading accuracy of about 0.005 mGal. Stations were located mainly to the east and west of São Miguel Island in order to cover the whole island. A special study was carried out in the Furnas area to make it possible to define the local structure, so gravimetric stations were more densely distributed in this zone (Figure 4). The altitudes of the stations were computed by means of known geodetic values, microbarometric leveling (with accuracy about 2 m) or geometric leveling (accuracy about 0.005 m).

For further terrain effect computations a topographic model of São Miguel Island and closed bathymetry (Figure 5) was constructed from 23436 points obtained by means of 1:25000 and 1:50000 topographic chart digitalization.

Gravity Anomalies and Anomalous Field Model

In order to obtain the distribution of refined Bouguer anomalies, we reduced the gravimetric data (the theoretical vertical gradient -0.3086 mGal/m was used for the free-air reduction) and calculated the terrain correction from the digital terrain model of São Miguel and nearby oceanic areas (up to 40 km from each station, because beyond this distance the correction can be considered to be uniform).

Ridley et al. [1974] worked with densities of 2600, 2300, and 2000 kg/m^3 for the gravity reduction in this area. In order to select one of these values, we calculated the refined Bouguer anomalies for those mass correction densities and then applied the process described below. The comparison of the results led us to choose 2300 kg/m^3 as the appropriate density value because for this value the adjusted density models and the short-wavelength gravity signals display less correlation with the topography. In our opinion, the value 1800 kg/m^3 applied by *Mendes and Gonçalves* [1980] to the Portuguese data is too low. With this adopted mass correction density, we obtained values for the topographic correction ranging between 0.8 and 14 mGal. Figure 6 shows the refined Bouguer anomalies obtained with this correction. The final accuracy that should be attributed to this map is not only induced by the measurement error and by the accuracy of the gravimeter; other conditioning factors must also be borne in mind. On the one hand, the use of microbarometric leveling to determine the altitudes of some of the stations results in a low degree of accuracy (of the order of 1.5 mGal) for the free-air reduction of those points. Furthermore, several of the stations are located in very rugged terrain, which means that the terrain correction might also be incomplete or incorrect. Therefore the stations with these type of problems will be affected by "noise" derived from these errors in addition to other effects caused by very local anomalies that are not very easy to interpret. The sum of these gravimetric signal perturbations is what we shall call "observational noise." The fact that this observational noise exists suggests that a process should be applied to reduce its immediate harmful effect on the stability of the gravimetric inversion process. On the other hand, regular geometric distribution of the data, though not necessary, does make the inversion process more regular. So, to construct a continuous model of the anomalous field, filtering the observational noise and interpolating the correlated signal, we applied a covariance analysis and a least squares prediction with the covariance function of the detected signal as was described by *Camacho et al.* [1994]. This process also allows us to determine the covariance matrix of the interpolated

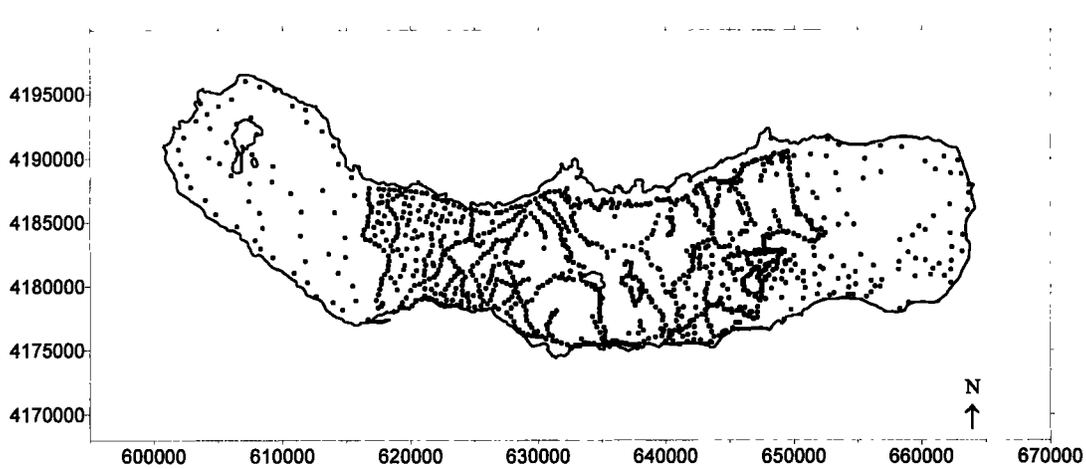

Figure 4. Distribution of gravimetric stations. The altitudes ranges from 2 m to 949 m.

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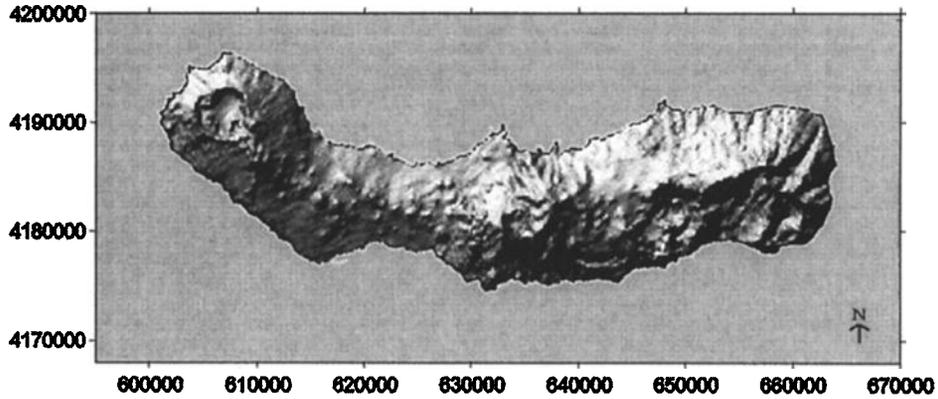

Figure 5. Terrain model of São Miguel Island obtained by 23,000 points from topographic charts. The highest elevations correspond to 1100 m. The caldera topography can be observed in this orthographic view.

anomalies, which will be both interesting and useful in the ensuing inversion process.

The observed anomaly values Δg_i, $i=1,\ldots,l$, where l is the number of stations, can be written in a vector as

$$\Delta \boldsymbol{g} = \boldsymbol{p} + \boldsymbol{v}$$

where $\boldsymbol{p}$ is a "systematic" component (in the sense of a mean value, a general slope or a very large wavelength component) and $\boldsymbol{v}$ is the correlated signal plus the observational noise. When the component $\boldsymbol{p}$ has been calculated, for example, by a polynomial fit, the residual values $\boldsymbol{v}$

$$\boldsymbol{v} = \Delta \boldsymbol{g} - \boldsymbol{p}$$

can be submitted to a covariance analysis to detect the first part of the correlatable signal, $\boldsymbol{s}$, of the anomalies. For this level of covariance, the noncorrelatable part $\boldsymbol{n}$ acts as the noise,

$$\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{n} \ .$$

For that purpose, assuming that signal covariance between two points, P_i P_j, depends only on the horizontal distance $d=\mathrm{dist}(P_i,P_j)$, i.e. $\mathrm{cov}(s(P_i),s(P_j)) = C(d)$, the correlation analysis of the $\boldsymbol{v}$ data gives us a distribution of empirical covariances. Further, these empirical covariances are adjusted by means of a suitable analytical function, $C(d)$, which allows us to calculate the values of the covariance matrix for the signal. *Barzaghi et al.* [1992] suggest various models of definite positive functions in such a way that there is a good fit to gravity data. From these models, in order to fit our covariances distribution, we selected the function given by

$$C(d) = a\ J_0(cd)\ e^{-bd}$$

where J_0 is the zero-order Bessel function and a, b, and c are parameters calculated so that the function $C(d)$ fits the empirical covariance values as best as possible.

Using the usual formulae for the least squares prediction [*Moritz*, 1980], the predicted signal $\boldsymbol{s}$ at the point P is obtained from the adjusted covariance function $C(d)$ and the data values at the observation points P_i as

$$\hat{\boldsymbol{s}} = \boldsymbol{C}_{Ps}\ (\ \boldsymbol{C}_{ss} + \boldsymbol{C}_{nn})^{-1}\ \boldsymbol{v}$$

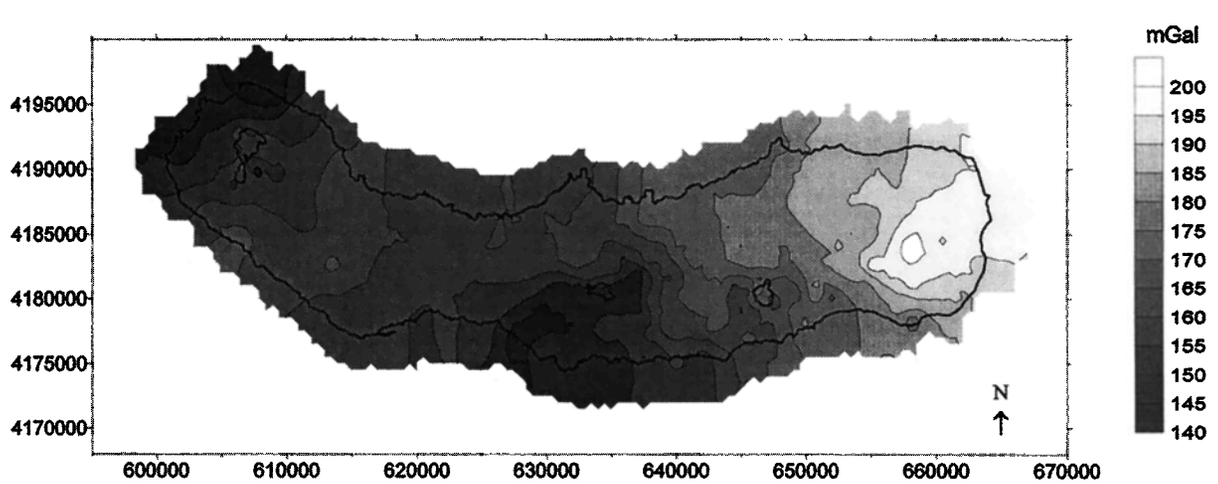

Figure 6. Map of refined Bouguer anomalies deduced from observation data. Contour interval of 5 mGal. Axis coordinates on Universal Transverse Mercator Projection system (m). The contours of the island and several lakes are added in this figure and the following figures.

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and the prediction error matrix for the adjusted signal is given by

$$E_{ss} = C_{PP} - C_{Ps} \, (C_{ss} + C_{nn})^{-1} \, C_{sP}$$

where C_{Ps}, C_{PP}, and C_{ss} are covariance matrix for the signal between the point P and/or the points P_i, and C_{nn} is a diagonal matrix for the noise variance (calculated from observations variance).

As described by *Camacho et al.* [1994], once the signal s is adjusted, we considered the resulting residual values $\Delta g - p - \hat{s}$ (noncorrelatable part for the previous covariance level) for a new covariance analysis to detect, if possible, a secondary correlated signal s'. This small secondary signal exists because the covariance functions adopted are established on the basis of a previously calculated optimum correlation step. If the gravimetric anomalies are caused by the superposition of effects of various wavelengths, the optimum correlation step may give evidence of the predominant signal, but it is too large to evaluate a very short wavelength correlatable signal. In a second covariance analysis of the previous residues, by obtaining a shorter optimum correlation step we will be able to identify and evaluate a smaller correlatable signal with a smaller "wavelength." The process

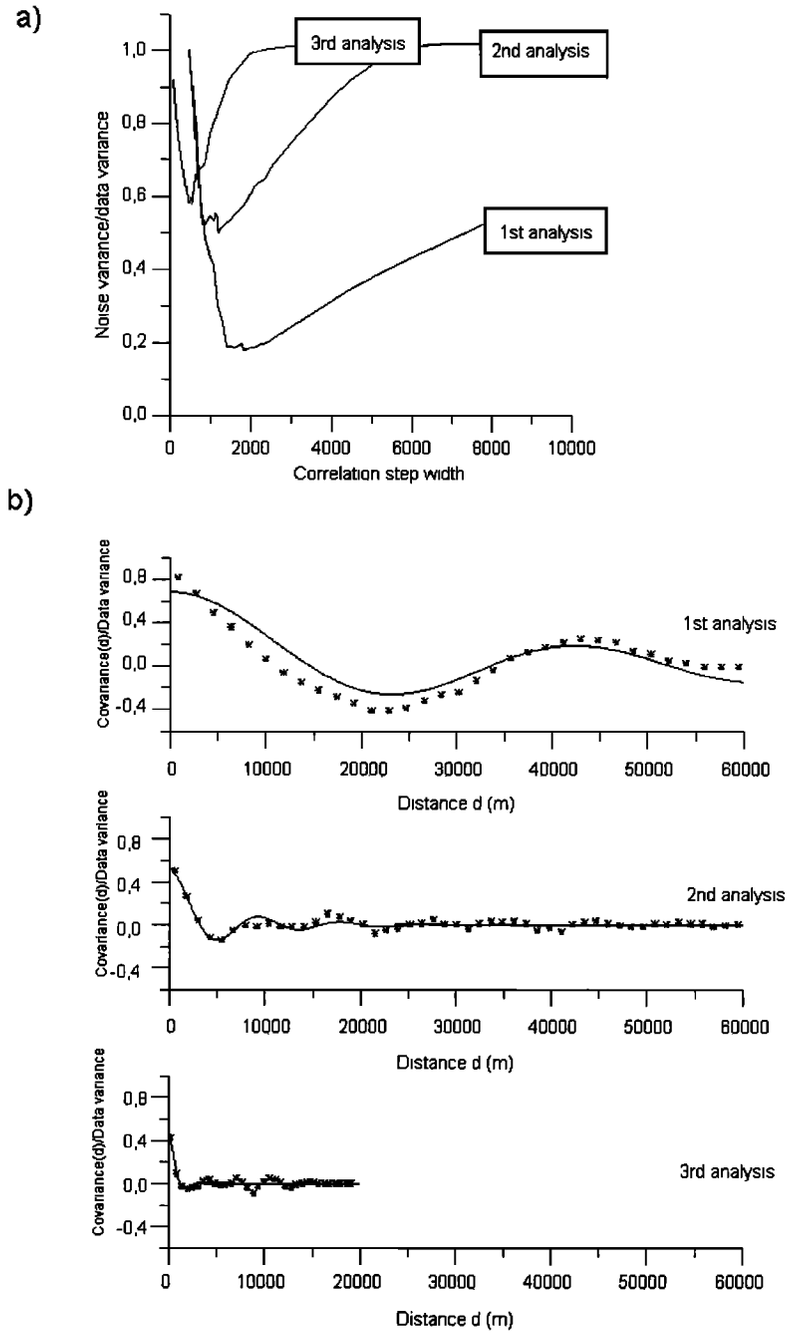

Figure 7. (a) Selection of the optimum correlation step, for each of the covariance analyses, as the step that minimizes the variance of the resulting noise. (b) Covariance functions adjusted (line) to the three successive empirical covariance distributions (stars).

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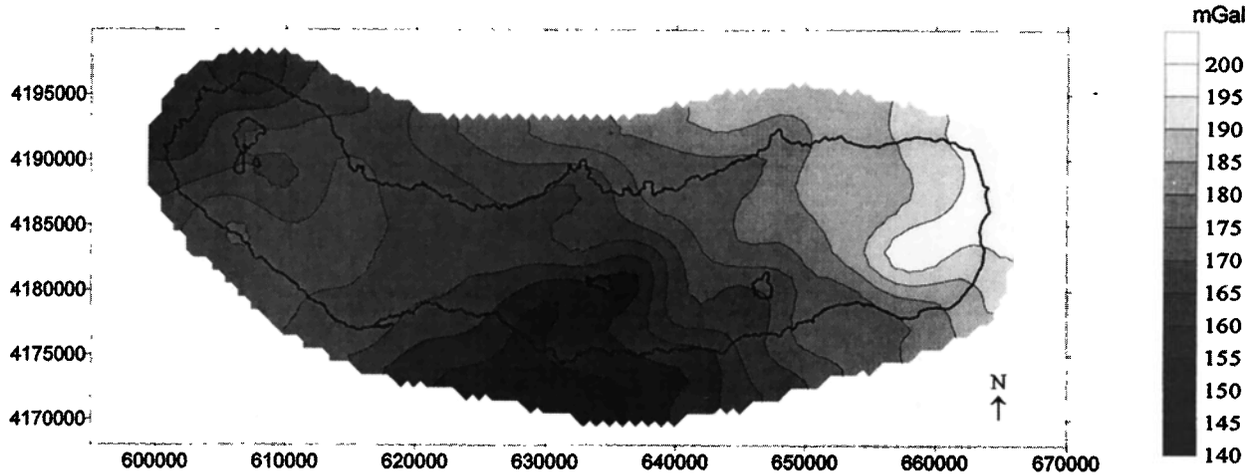

Figure 8. Least squares prediction model of the refined Bouguer anomaly field. This model represents the optimal linear interpolation, filtering the observational noise. Contour interval of 5 mGal.

is repeated until all the correlatable contents of the observed anomalies have been extracted. The final anomaly model is obtained by adding the signal components.

This process was applied to the São Miguel gravity data. In the first adjustment, a polynomial surface of degree 1 was determined as a systematic component, p. In the subsequent covariance analysis (see Figure 7) we detected two levels of correlated signal for the whole island and another for the specific zone of Furnas. To avoid the disturbance on the covariance analysis because of the unevenly distribution of data set, we picked out a data selection, as homogeneous as possible, from the whole stations. This choice was applied for the polynomial fit and the first covariance analysis, selecting stations with a minimum mutual distance of 800 m and 400 m, respectively. Figure 8 shows the anomaly model obtained for the island, and Figure 9 is a higher-resolution view for the Furnas area. Figure 10 is the error map for this gravity prediction model, with a mean value of 0.9 mGal. It is easy to see the relationship that exists between the abundance of stations and the error in the prediction of the anomalies. Finally, Figure 11 shows the distribution of final residues of the filtering, which would result from irregularities in the gravimetric reduction process and to very local anomalies (observational noise). The residuals are centered around zero and have a root mean square error of 1.1 mGal.

Gravity Inversion

To obtain information on a density contrast structure from the gravity anomalies, we used a linear discrete 3-D inversion treatment that only includes mathematical hypothesis. We divided the island's subsoil into a finite three-dimensional grid of n rectangular (cubic) prisms. Then we considered a global adjustment with m equations, corresponding to

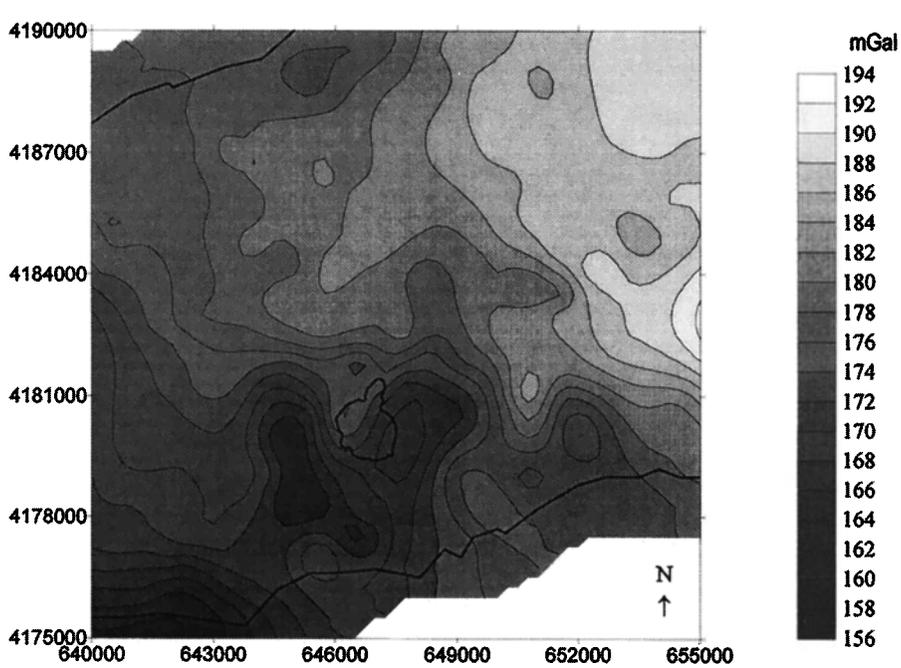

Figure 9. Least squares prediction model of refined Bouguer anomaly field for the Furnas area. A higher-resolution view. Contour interval of 2 mGal.

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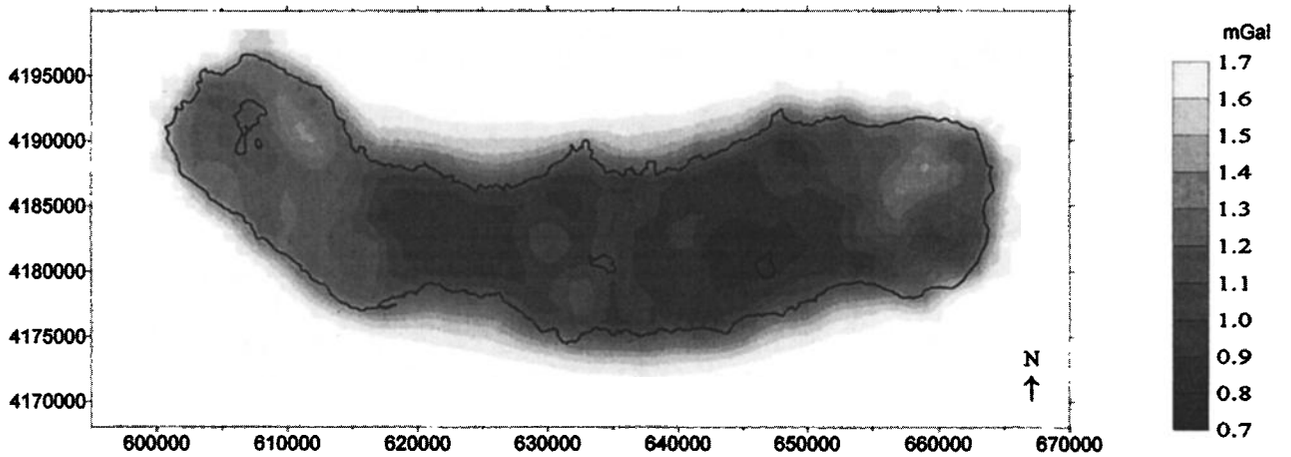

Figure 10. Map of the anomaly prediction error. Contour interval of 0.1 mGal. A dependence of stations distribution is observed on the error map.

anomaly model points (as was predicted in the previous section), and n unknown densities, corresponding to the prisms. The "observation" equation system has a linear tructure:

$$A\ x = t_{obs} \tag{1}$$

where x is the n vector of unknown anomalous densities ρ, t_{obs} is the m vector of "observed" data (the predicted anomalies Δg) and A the design (m,n) matrix. Unlike the traditional use of the least squares adjustment, in these inversion problems as much as possible unknowns are wanted to conclude from the existing observations, and therefore usually n > m. Then, a rather different formulation is desired. The (i,j) element of the design matrix A corresponds to the gravitational attraction of the prism j, for density $\rho = 1$, over the observational point $P_i(x_i,y_i,z_i)$, and is given by *Pick et al.* [1973] [see also *Nagy*, 1966]:

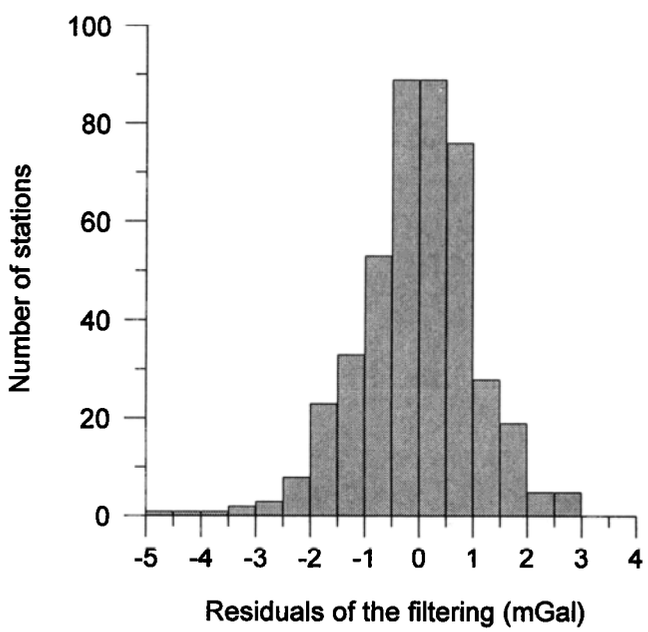

Figure 11. Distribution of the final residues of the filtering. The corresponding standard deviation is of the order of 1.1 mGal.

$$a_{ij} = -G\rho \left[\left[\left[x \ln(y+(x^2+y^2+z^2)^{1/2}) + y \ln(x+(x^2+y^2+z^2)^{1/2}) + z \arctan \frac{z(x^2+y^2+z^2)^{1/2}}{x\ y} \right]_{u_1-x_i}^{u_2-x_i} \right]_{v_1-y_i}^{v_2-y_i} \right]_{w_1-z_i}^{w_2-z_i} \tag{2}$$

where G is the gravitation constant, density $\rho = 1$, the edges of the j prism are parallel to the reference axes, and the limiting coordinates for its volume are u_1, u_2 for x coordinate, v_1, v_2 for y coordinate, and w_1, w_2 for z.

As common with potential fields inversion, the resolution of the gravimetric inversion has a substantial problem of nonuniqueness. Some additional information is necessary to avoid this nonuniqueness problem. A general way of including the information is obtained by probability density distributions [*Tarantola*, 1987]. In the case of a Gaussian distribution hypothesis, its central value and the covariance operator describing the uncertainties contain the information and let us apply a least squares treatment. A Gaussian predetermined distribution hypothesis (given by the corresponding covariance matrix) on the unknown parameters solves the nonuniqueness problem. We suppose that the a priori information on x may be described by the a priori value x_0, with Gaussian uncertainties described by the covariance matrix C_M. We also suppose that the predicted values t_{obs} have a Gaussian uncertainty described by the covariance matrix C_d. Finally, we suppose that the forward relation (1) holds only approximately with Gaussian uncertainties given by the covariance matrix C_T. Thus the posterior parameter estimation is Gaussian with center [*Tarantola*, 1987]:

$$\hat{x} = x_0 + C_M A^t (A\ C_M A^t + C_D)^{-1} (t_{obs} - A\ x_0)$$

and covariance matrix

$$C_{M'} = (A^t\ C^{-1}{}_D\ A + C^{-1}{}_M)^{-1}$$

where

$$C_D = C_d + C_T\ .$$

Generally, matrixes C_d and C_T are not well known. We can use the error matrix of the anomalies prediction as the

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covariance matrix C_d for the data parameter,

$$C_d = E_{ss} .$$

Furthermore, if the theoretical forward relationship given by (1) and (2) is exact, then, we could suppose

$$C_T = 0 \quad , \quad C_D = C_d .$$

Nevertheless, the prismatic partition of the subsoil volume may be a poor discretization (too thick a partition) for the distribution of densities, and therefore part of the correlated anomalies will not be well absorbed by the discrete solution. That portion of the anomalies must be assumed as part of the inversion uncertainties. The data covariance for the inversion must be bigger than the one calculated previously by means of covariance analysis. Applying the usual techniques, we introduce a positive factor λ^2 to regulate the mutual fitness between the data covariance influence and the parameter covariance influence [*Marquardt*, 1970; *Richardson and MacInnes*, 1989; *Burkhard and Jackson*, 1976]:

$$C_D = \lambda^2 C_d .$$

With this damping factor, the adjusted values for the parameters (prisms densities) are

$$\hat{x} = x_0 + C_M A^t (A C_M A^t + \lambda^2 C_d)^{-1} (t_{obs} - A x_0) \qquad (3)$$

which correspond to the maximum likelihood point defined by

$$(x-x_0)^t C_M^{-1} (x-x_0) + \lambda^2 (Ax - t_{obs})^t C_d^{-1} (Ax - t_{obs}) = \min$$

Therefore the positive factor λ^2 allows us to regulate the suitable balance between the data fit level and the adjusted structure smoothness.

For an initial study without previous specific information about the model parameters (prism densities), we can take as a priori value $x_0 = 0$ and a model covariance matrix C_M given by a diagonal normalizing matrix of nonnull elements the same as the diagonal elements of

$$A^t C_d^{-1} A$$

corresponding to weighted mean squared attraction of each model prism upon the survey area.

For the purposes of practical implementation, we predict anomaly and error values for points equally spaced (though not necessarily) over a region limited by an extreme error value. The gridding step width is a rounded value of the step width calculated for the gravity anomaly covariance analysis (such as that producing a minimum of the noise variance). Furthermore, a prism distribution is selected with a side close to the data step width. We can then calculate the values (2) of the elements of the design matrix A. The model region is limited by a maximum of the model variance values as the corresponding elements of the diagonal matrix C_M.

The gravity anomalies are relative values, and therefore we can subtract an offset constant value G_{off}, obtaining rather different structural models. For this reason, to produce an optimal adjustment we select an offset value (such as the simplified regional trend) that produces a minimum of the further mean (weighted) residual value. That is to say, we look for a weighted residual distribution of null mean value. The choice of a suitable value of λ is tentatively made. We consider the resulting residual values weighted by their corresponding covariance matrix C_d, and we call their mean value v_{mean}. For each selected λ value, a fixed level of data t_{obs} variation produces a level of variation for the (weighted) residual mean value v_{mean}. Figure 12 shows, for a fixed increment of the offset constant G_{off}, the variation of the value v_{mean} of several λ values. A characteristic minimum is observed, and that λ value is chosen as optimal.

Finally, if we know the realistic maximum and minimum limits for the model density contrast, they can be applied as follows: Once the direct solution is obtained, the resulting model can be modified by keeping the density values contained between the limits and by setting the exceeding density values just equal to the corresponding contrast limits. This modified model can be used as an a priori model for a new inversion. Clearly, we cannot use the same λ value for this second inversion (now we want the same adjustment level but only with small model modifications). If we refer to the solution first obtained from the a priori model x_0 (for us $x_0=0$) as x_1, we define the vectors

$$b_0 = A x_0 - t_{obs} \quad , \quad b_1 = A x_1 - t_{obs} .$$

Then, if we call the values corresponding to the initial and further inversions λ_0 and λ_1, we take

$$\lambda_1^2 = \lambda_0^2 \frac{b_0^t C_d^{-1} b_0}{b_1^t C_d^{-1} b_1} .$$

The new values, x_1, λ_1, b_1 for the parameters are used in (3) to obtain a new approximation, and so on to obtain a model with density contrast limited by the fixed values.

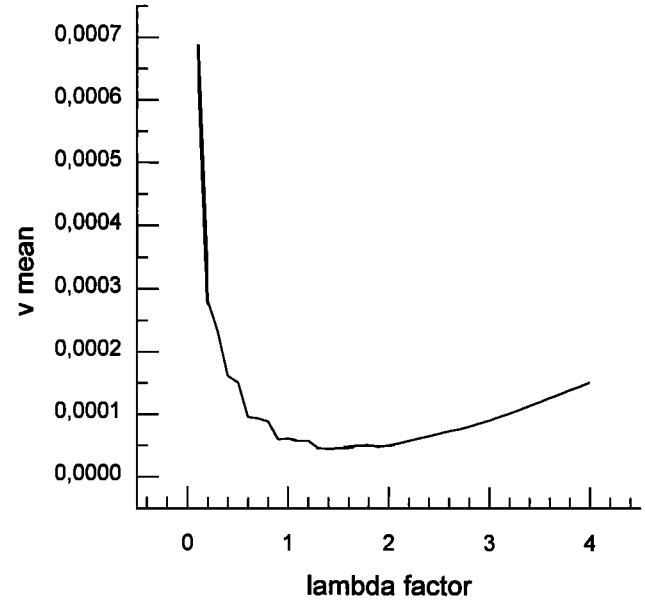

Figure 12. Variation of the mean residual value v_{mean} for several λ values. A characteristic minimum is observed that lets us choose an optimum value for this control parameter.

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This method was first applied for the São Miguel global anomaly field. A data grid with a step of 1500 m was adopted from our model of anomalies, giving a total of 480 equations. Furthermore, the subsurface portion was composed of cubic prisms with a side of 1650 m, reaching a volume with the bottom located at a depth of 25,000 m (in accordance with the characteristics of our grid and the calculation capacity). Then a total of 11,100 unknown parameters were considered (close to our RAM limit, in an ALPHA 2100 VMS computer). Finally, a limiting value of -270 kg/m^3 and 240 kg/m^3 for the resulting density contrast was adopted, taking into consideration the extreme values that resulted initially in the central region of the unknown volume and in accordance with the maximum contrast values foreseeable a priori (according to *Talwani et al.* [1965]). The 3-D anomalous density model that results is shown by means of several horizontal sections and vertical profiles (Plate 1).

The same method was applied for the more detailed Furnas gravity anomalies to obtain a particular, higher-resolution model of anomalous density structure. For this adjustment the following values were adopted: gridding step of 400 m, prisms with sides of 440 m, 620 equations and 8300 unknown parameters. The extreme density contrast values adopted for the readjustment were -520 kg/m^3 and 330 kg/m^3 from a shallower model with a higher resolution

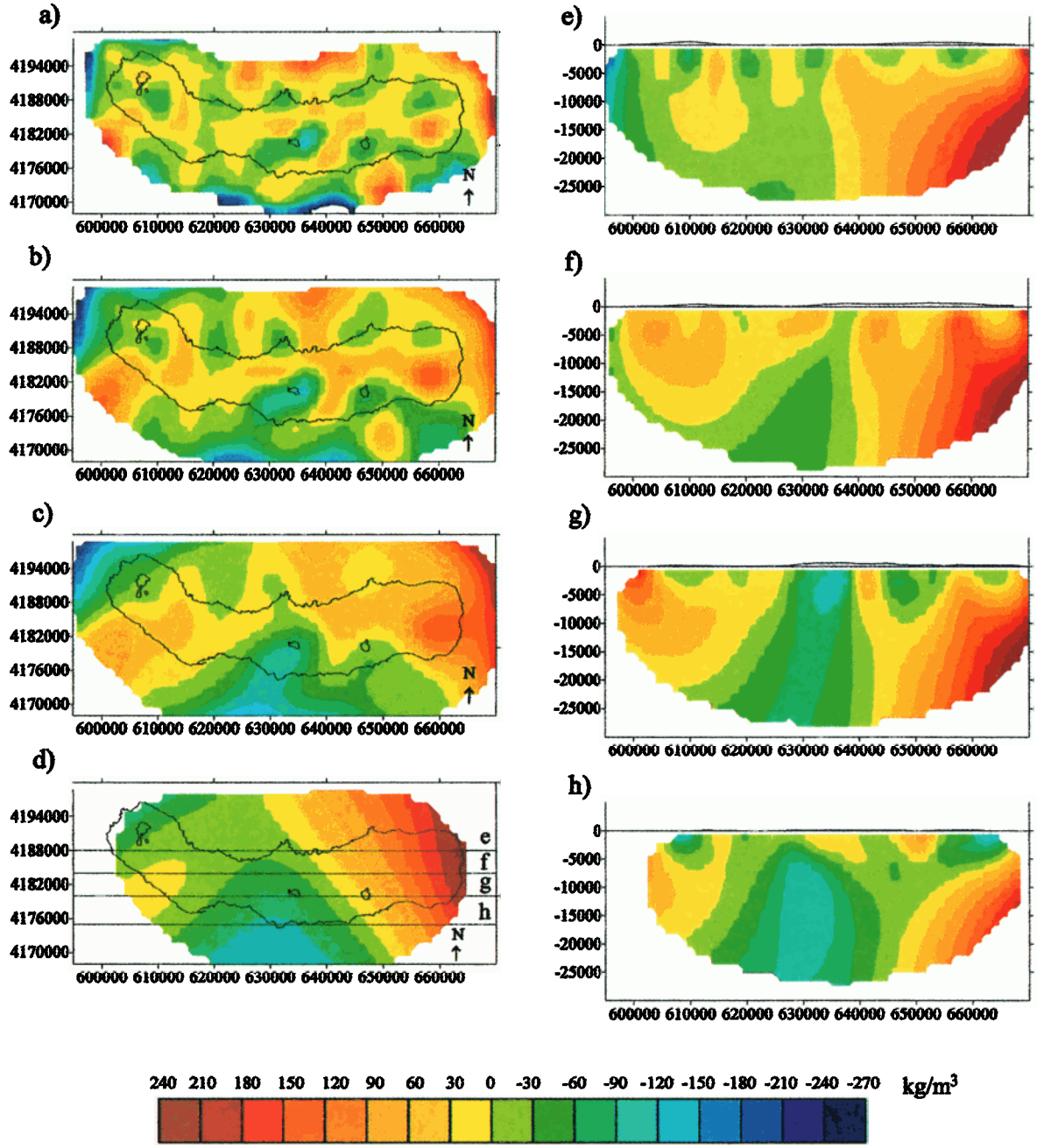

Plate 1. Adjusted model of anomalous density contrast of São Miguel Island obtained from the gravimetric inversion. Horizontal sections: (a) z=-1000 m, (b) z=-4000 m, (c) z=-8000 m, (d) z=-20,000 m. Vertical W-E profiles (indicated in Plate 1 d): (e) Y=4,188,000, (f) Y=4,184,000, (g) Y=4,180,000, (h) Y=4,175,000. Contour interval of 30 kg/m^3.

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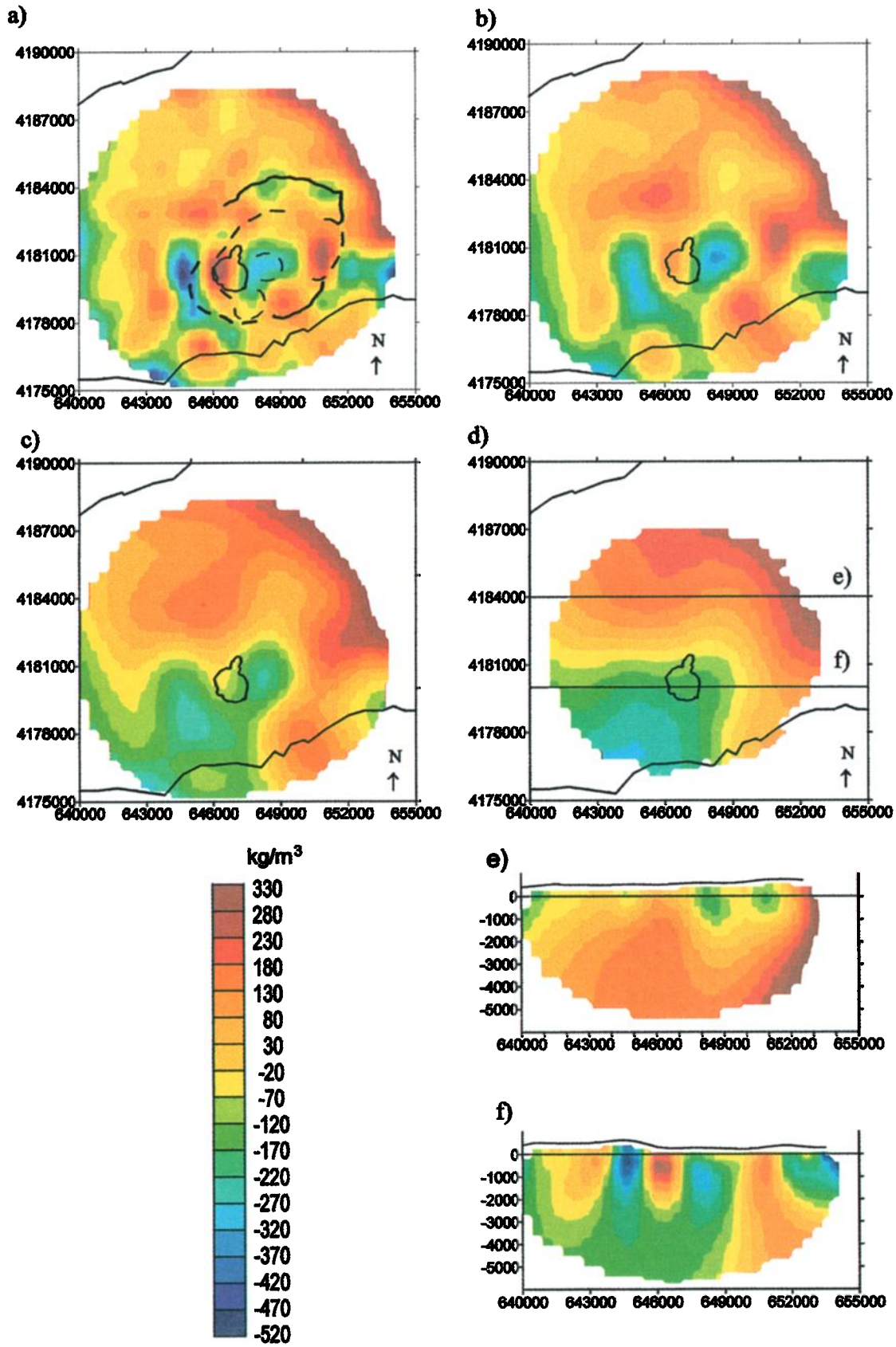

Plate 2. Adjusted model of anomalous density contrast of the Furnas area obtained from the local gravimetric inversion. Horizontal sections: (a) z=-500 m, (b) z=-1500 m, (c) z=-2500 m, (d) z=-4000 m. Vertical W-E profiles (indicated in Plate 2 d): (e) Y=4,184,000, (f) Y=4,180,000. Contour interval of 50 kg/m^3.

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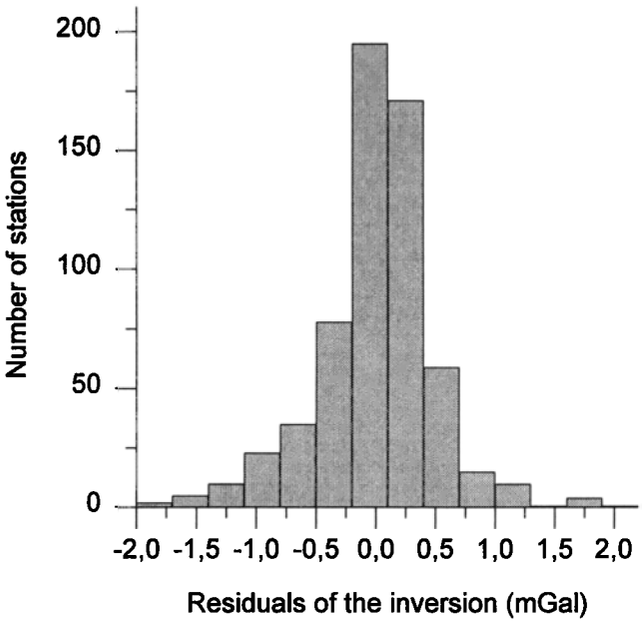

Figure 13. Distribution of the inversion model residuals, obtained as the difference between the anomalies calculated for the densities model and the filtered anomalies. The result is a mean value of null and a standard deviation of 0.6 mGal.

than the model calculated for the island and in which the surface volcanic matter is responsible for most of the anomalies. This adjusted model is shown by means of several horizontal sections and vertical profiles (Plate 2).

The extent to which the anomalies produced by this model fit the data anomalies (Figure 13) depends essentially on the value of parameter λ^2. For the value corresponding to the aforementioned optimization criterion, the result is centered residues with mean square errors of 1.5 mGal and 0.6 mGal for the anomalies of the whole island and Furnas, respectively (these values are similar to the mean error estimations for the least square prediction values of the anomalies). The relative accuracy of the adjusted density values for the unknown prisms drops heavily the farther down the blocks are located and in relation to the peripheral situation. Figure 14 is a vertical profile of the distribution of relative accuracies for the densities of the unknown prisms. With this inversion process the models are well-defined in the surface layers, whereas at lower depths the mass anomalies are rather "diluted." Moreover, the peripheral prisms, which are therefore more imprecise, may accumulate regional effects that do not match the unknown volume considered. Furthermore, we must take into account that the gravity observation is not sensitive to horizontal stratification, so this model must be considered as anomalous masses with regard to a mean subsurface structure, where an additional stratification from other geophysical techniques (mainly seismology) could be added.

Most Relevant Characteristics of the Adjusted Model: Conclusions

The Azores spreading center that resulted in the formation of the archipelago may be considered as the most relevant tectonic characteristic of the deep structure of Sao Miguel Island [*Krause and Watkins*, 1970]. From the gravimetric point of view, these types of structures tend to be associated with density minimums. Thus, *Talwani et al.* [1965], for instance, suggest a low-density lenticular cross-sectioned structure located at the top of an anomalous mantle. This characteristic feature of the zone is also observed in the density model that we have adjusted as the most notorious element at great depths.

In the deep horizontal sections of the model (e.g., 20 km), two main minimums are observed in the area of the mantle (which *Ridley et al.* [1974] locate below 14 km). These minimums are closely related to the tectonic pattern of the Terceira Rift, marking an azimuth of 120° for the spreading center under the island, as already suggested by *Searle* [1980]. At greater depths (the model goes as far down as 25 km, as shown in the vertical profiles in Plate 1), the minimums tend to join up, emphasizing the position of the spreading. It is below 20 km that these minimums display a well-defined geometry with NW-SE and NE-SW ridges, which suggest the existence of tectonic fracture lines related to the sides of the rifted basins and the ridge ranges that flank them, associated with the spreading center. As we approach the surface (at approximately 8 km), these two global minimums tend to disappear and are limited to the area of Fogo and Furnas, where they trend NE-SW. Likewise, the minimum persists in the Mosteiros Graben area, where *Guest et al.* [1994] suggest the presence of active faults.

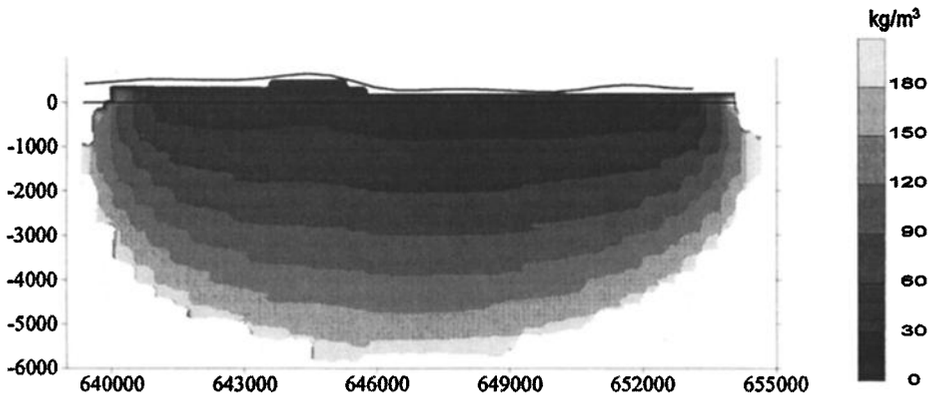

Figure 14. Vertical profile of the subsequent distribution of relative errors for the prisms resulting from the inversion process.

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At shallower depths (4 km) other minimums are also observed in the areas of Povoaçao and Sete Cidades, the latter coinciding with the area of the historic eruption (see Figure 3). These minimums at intermediate depths may be associated with areas of fissures, hydrothermal alterations or with low-density magmatic chambers, perhaps containing acid or intermediate differences. With regard to Sete Cidades, Fogo, and Furnas, several authors [*Machado*, 1973; *Forjaz*, 1996; *Sigmundsson et al.*, 1995] place the chambers at depths of around 4 or 5 km, which is consistent with our model. The minimum located in Fogo, an area of peculiar geothermal characteristics, is the most notable feature at the shallowest level (1 km) of the model.

As regards the density maximums in the model, the main maximum is located at a depth of more than 12 km in the Nordeste shield (northeastern zone). Other more local maximums appears at shallower depth (4 km) in Sete Cidades, in Congro, in Barrosa (NW of Fogo), in Serra Gorda, and especially in the Serra da Tronqueira to the north of Povoaçao (Figure 3 shows these locations). These maximums may correspond to old basaltic shields and to partly solidified magmatic bodies. In Congro, the maximums clearly mark the WSW-ENE tendency put forward by *Guest et al.* [1994] as one of the structural characteristics.

Let us now consider the anomalous masses model resulting from the adjustment for Furnas. The most notable element is the minimum, centered below the Pico do Gáspar crater, site of one of the historic eruptions of Furnas (1440). This minimum could be centered 1.0 to 1.5 km deep. For this type of volcano the minimum can be attributed to an uncompacted silicic caldera infill resulting from acid explosive eruption and collapse processes [*Rymer and Brown*, 1986]. Another interesting minimum is to be found to the west of lake Furnas, on the edge which was the expansion of the caldera as a result of the largest eruption of the zone in the last 5000 years [*Guest et al.*, 1994]. This minimum is of an elongated shape in a SE-NW trend, suggesting that its origin was an infill by light materials from a fracture zone. Both the minimum density surface structures of Furnas have a common substratum in a less developed area of low density. This zone, which extends toward the deep minimums typical of the island's structure, may be attributed in this type of volcanoes to the possible contribution from low-density magma bodies about 4-5 km depth.

Other, less important surface minimums are to be found in the profiles. Most of the volcanic episodes occurred inside the caldera or nearby. Nonetheless, there have been Strombolian-type eruptions outside the edge of the caldera, on the Northern flank, and their relationship to the center of Furnas is not clear [*Guest et al.*, 1994]. It is this zone, the northern wall of the first caldera, that contains the W-E direction minimums that could be associated with this volcanism. The vertical profiles demonstrate that these minimums are independent of the minimums related to other events. Furthermore, the minimum located to the east, outside the caldera walls, could be associated with the eruptions of the neighboring volcanic complex of Povoaçao.

Moreover, the anomalous density maximum of the characteristic basaltic shield of Serra da Tronqueira on the NE edge of island slopes down to the SW, toward the base of Furnas, where it appears as a strong maximum. Another anomalous density maximum of note is the one located, at surface level, on the west of the Furnas Lake crater, which was the result of a prehistoric eruption. Its nature may be attributed to an isolated, more basaltic episode that contrasts with the less compacted materials of its environment from eruptions.

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